

ABSTRACT

Cardiovascular Diseases (CVDs) continue as a primary killer of premature death globally making cost-effective and intelligent monitoring solutions that can be used outside of hospital settings essential.

In this project related to Real-Time Cardiovascular Event Detection and Prediction system. It uses multi-parameter physiological sensor, machine learning, and IoT. The hardware platform is an ESP32 microcontroller with an AD8232 ECG sensor, MAX30102 pulse oximetry and heart rate sensor (SpO₂), and an LM35 temperature sensor. All these sensors are collecting important bio-physiological parameters such as ECG, SpO₂, BPM, and body temperature. There is no direct integration in the prediction model, the hardware unit works independent of the prediction model and have dedicated sensing and display module.

A sensor value is displayed on the LCD screen and is manually entered by a healthcare professional or the operator into the web dashboard (based on Flask). These inputs are then fed to a trained machine learning pipeline that consists of four supervised classification algorithms: Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbour (KNN) and Decision Tree (DT) trained on the Cleveland Heart Disease dataset. The Random Forest (RF) classifier gave the highest accuracy of 93.18% better than SVM (89.5%), KNN (87.2%) and Decision Tree (85.6%), and it was selected as prediction model.

Based on the manually entered values, the system gives an immediate risk classification, either "Heart Disease Detected" or "No Heart Disease Detected" and provides some performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix visualization.

It offers a scalable, non-invasive, and cost-effective alternative to the traditional diagnostics, and has a great potential to be used in remote healthcare, telemedicine applications.

Keywords: Cardiovascular Disease Detection, Sensor Fusion, Machine Learning, ECG, SpO₂, Random Forest, SVM, KNN, Decision Tree, ESP32, IoT, Real-Time Monitoring, Heart Risk Prediction.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CVD	Cardiovascular Disease
ECG / EKG	Electrocardiogram
SpO ₂	Peripheral Oxygen Saturation
BPM	Beats Per Minute
ML	Machine Learning
RF	Random Forest
SVM	Support Vector Machine
KNN	K-Nearest Neighbour
DT	Decision Tree
IoT	Internet of Things
ESP32	Espressif Systems MCU with Wi-Fi & Bluetooth
MCU	Microcontroller Unit
AD8232	Analog Devices ECG Signal Conditioning IC
MAX30102	Maxim Integrated Pulse Oximetry Sensor
LM35	Linear Temperature Sensor (National Semiconductor)
LCD	Liquid Crystal Display
API	Application Programming Interface
TPR	True Positive Rate
FPR	False Positive Rate
F1	F1-Score (Harmonic Mean of Precision and Recall)
AI	Artificial Intelligence
SVM	Support Vector Machine
UCI	University of California, Irvine (Dataset Repository)
AICTE	All India Council for Technical Education
NAAC	National Assessment and Accreditation Council
NBA	National Board of Accreditation

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Cardiovascular diseases (CVDs) are an umbrella term for all diseases related to the heart and blood vessels, such as coronary artery disease, arrhythmias, heart failure and sudden cardiac arrest. As per World Health Organization (WHO), every year about 17.9 million people die from cardiovascular disease (CVDs) which roughly makes up 32% of all deaths worldwide. The burden of cardiovascular disease (CVD) has increased significantly in India in the last 2 decades, mainly because of urbanization and sedentary lifestyle, changes in dietary habits, psychological stress and rising prevalence of diabetes and hypertension. The financial burden is also substantial, with cardiac hospitalizations, surgeries, and the ongoing expenses of medication creating a significant burden in the health care system and on individuals, particularly when taken into account over a prolonged period.

In addition to treating these patients, the objective in cardiovascular disease management is the early detection and ongoing monitoring of these patients. Many cardiac incidents — such as heart attacks, arrhythmias and sudden cardiac death — happen without warning in people who were once thought to be healthy. The need to have intelligent, continuous, and proactive health monitoring systems that can identify anomalies before they arrive at life-threatening levels is evident. Early warning signs could be indicated by physiological markers like ECG patterns, blood oxygen saturation, heart rate, and body temperature, which could be crucial for timely medical intervention and save lives.

Typical forms of cardiovascular assessment that are used in the clinic are periodic visits to a hospital, resting ECG measurements, echocardiogram, blood cholesterol analysis and stress testing. These techniques are accurate but sporadic. They only give a short window to an individual's cardiac health, and they are not really capable of detecting irregularities that may occur occasionally or during the normal course of life. Moreover, these hospital-based interventions are costly, time-consuming and difficult to access for patients in rural and remote regions, where specialist cardiology services are still significantly lacking, or even non-existent. As sensor technologies and algorithms that can perform machine learning analyses become more

and more connected to the Internet of Things, low cost, compact microcontrollers become more and more widely available and more and more powerful, new opportunities for the development of intelligent, point-of-care, and outside-the-clinical-lab diagnostic technologies emerge. The present project aims to develop a system that continuously measures and displays important physiological parameters using an LCD screen, with an ESP32 microcontroller connected to an AD8232 ECG sensor, an MAX30102 SpO2 and heart rate sensor and an LM35 temperature sensor. This piece of equipment provides a convenient, mobile measurement location – allowing a healthcare professional or trained user to view live sensor data on the unit's display. These values are then manually input into a Flask based web application that has a trained machine learning model using the Cleveland Heart Disease dataset to classify immediate cardiovascular risk.

This hybrid system that integrates hardware level sensor with software level intelligence is a middle ground between continuous physiological monitoring and intelligent clinical decision support. It eliminates the need for expensive integrated diagnostic equipment and yet still harnesses the predictive capabilities of supervised machine learning algorithms. In primary healthcare settings, rural clinics and community health camps, where time, cost and relatively good accuracy of cardiac screening is crucial to the patient's care, the system proves highly effective.

1.2 PROBLEM STATEMENT

Although clinical diagnostic tools are widely available, there is a critical need for intelligent cardiovascular monitoring which is accessible, affordable and feasible – especially outside the walls of high-tech hospitals. Current portable monitors and wearable devices are only able to capture one physiological measure, for example, heart rate alone, and have few ability to classify or predict cardiac risk with any real meaning. Fewer physiological assessment results means more false alarms, less diagnostic confidence, and the inability to pick up the interdependent, complex nature of cardiovascular conditions.

In addition, fully automated and wirelessly integrated diagnostic systems, although technically achievable, is a very complex hardware system, costlier to develop and requires a strong communication system, which are difficult to deploy in resource-limited or remote healthcare settings. Thus, an efficient, cost-effective, and non-invasive solution that balances simplicity of hardware with intelligence of diagnosis is needed.

To fill this gap, the proposed system has a deliberate 2-stage architecture: the embedded system

consists of ESP32 microcontroller connected to an AD8232 (ECG sensor), a MAX30102 (SpO2 and heart rate sensor) and an LM35 (temperature sensor) which continuously collects and displays real-time physiological data on an LCD display. These values are then manually entered into a web application that is implemented in Flask and which immediately classes a user's cardio risk using a validated machine learning model. This method avoids the costly process of directly integrating hardware with the model and maintains the predictive power and clinical relevance of a supervised machine learning pipeline used for the Cleveland Heart Disease dataset. The outcome is a system that can be deployed, interpreted and accessible to healthcare workers.

1.3 PROJECT OBJECTIVES

This project has the following main aims:

- To design and develop an IoT based Cardiovascular event detection and prediction system that utilizes supervised machine learning algorithm and Embedded hardware for intelligent risk classification.
- Using biomedical sensors and interfacing with an ESP32 microcontroller to measure different physiological parameters like ECG waveform, blood oxygen saturation (SpO2), heart rate (BPM) and body temperature and then displaying all of these parameters in real-time on an LCD screen.
- To create an operator-assisted workflow where a healthcare professional or trained technician are responsible for observing the sensor values, manually entering them in a web-based prediction interface, which is used for machine learning-based analysis.
- To design a complete machine learning pipeline for four supervised classification algorithms namely Random Forest, Support Vector Machine (SVM), K-Nearest Neighbour (KNN), Decision Tree, trained and validated on Cleveland Heart Disease Dataset.
- For improving the predictive capability of each classification model, they will be preprocessed through the techniques of normalization, feature scaling and feature importance analysis.
- To compare and contrast all four models based on standard performance metrics (accuracy, precision, recall, F1-score and confusion matrix) and to determine which of these algorithms is the best performer for deployment.

- To develop and deploy the best performing machine learning model onto a web dashboard using flask with the ability to accept manually entered physiological readings as input and classify the patient as having "Heart Disease Detected" or "No Heart Disease Detected" as output.
- To visualize accuracy curves, confusion matrices, and show comparisons of metrics for the models in the web interface, which will aid in clear clinical decision making.
- To provide a scalable, non-invasive, cost effective diagnostic solution that has real-world potential for implementation in primary healthcare facilities, rural clinics and telemedicine.

1.4 SCOPE OF THE PROJECT

The project is limited in scope to cardiovascular event detection and physiological sensor data-based risk prediction via machine learning:

- Three biomedical sensors AD8232 (ECG), MAX30102 (SpO2/HR) and LM35 (Temperature) are interfaced with the ESP32-based embedded system. The hardware unit is a complete sensor and display unit that can be used as a stand-alone system, collecting and displaying real-time physiological data on an LCD.
- The system is built on a human-in-the-loop, practical workflow. In environments where continuous wireless data transmission infrastructure may not be available, the physiological values shown on the hardware LCD are observed by a trained health care professional or technician, and manually entered in the web-based prediction interface.
- The project involves creating a machine learning pipeline with the Python programming language, scikit-learn for training, evaluating, and deploying classification models on the Cleveland Heart Disease dataset, and Flask for exposing the model to users. The web dashboard offers the prediction interface, real-time risk output, model performance visualizations such as confusion matrices and graphs for comparison of metrics.
- Physiological readings were manually entered as input data and a classification system was developed to determine if the patient had heart disease or not, with accuracy and confidence metrics (accuracy, precision, recall, F1-score) transparent and meaningful for clinical users.
- The machine learning models have been trained and validated on the Cleveland Heart

Disease dataset, a widely used dataset in medical AI studies that includes 14 clinically relevant features. The prediction scope is thus limited to risk factors and demographic patterns included in this set. Automatic data acquisition from ESP32 directly to the web application, cloud integration for longitudinal patient monitoring, and mobile health platform support, extension to other medical conditions like diabetes, stroke etc. are identified as future scope of the framework. The system is targeted to the practical deployment in the primary healthcare centers, rural clinics and community health screening camps where low-cost and accessible cardiac screening is most desperately required.

- The design was aimed at being easy to use by everyday healthcare workers, with the design being applicable to use in the real world without the need for high tech infrastructure.

1.5 EXISTING SYSTEM

1.5.1 OVERVIEW OF THE EXISTING SYSTEM

Most ways to check for heart problems still happen in hospitals with doctors doing tests every so often. They use machines to get things like ECG readings and blood pressure and cholesterol levels. Then the doctor looks at all that to figure out the risk. It only gives a quick look at how the heart is doing at that moment though.

Some people have tried making computer models to predict risks using data from places like the Cleveland set. They use things like logistic regression or decision trees and nearest neighbors to sort patients into groups. But these are mostly just for studies and do not connect to any actual devices or sensors that could check things live. It seems like that leaves a big gap when you want something that works outside of research papers.

IoT setups that exist usually just track one thing at a time like heart rate or oxygen levels. They do not pull together different sensors the way you would need for a full check on cardiovascular stuff. A lot of them do not even have good prediction from machine learning and just show the numbers or send alerts. That makes them hard to use in places with fewer resources and I think the missing interface for regular staff makes it worse. Maybe the whole approach needs more.

1.5.2 BLOCK DIAGRAM OF EXISTING SYSTEM

The general workflow of the existing cardiovascular monitoring and prediction system, as illustrated in the block diagram above, follows a sequential pipeline from raw dataset input through to final prediction results. The key stages of this existing workflow are described below:

- **Stage 1 – Dataset Input and Data Collection:** The existing system begins with a static, pre-collected clinical dataset — such as the Framingham Heart Study (FHS) Dataset — as its primary data source. Unlike real-time physiological acquisition, this approach relies entirely on historical patient records gathered through hospital-based laboratory tests and clinical evaluations. Demographic and physiological attributes such as age, sex, resting blood pressure, serum cholesterol, fasting blood sugar, and ECG findings are recorded manually by clinical staff during scheduled patient visits.
- **Stage 2 – Data Preprocessing:** The collected dataset undergoes a data preprocessing stage to handle missing values, remove inconsistencies, and prepare the data for further analysis. This step is performed offline using data processing tools and scripts, with no real-time or continuous data feed involved.
- **Stage 3 – Feature Selection and Elimination:** Relevant clinical attributes are identified and irrelevant or redundant features are eliminated using statistical methods. This stage aims to reduce dimensionality and improve the signal-to-noise ratio within the dataset, retaining only those attributes that contribute meaningfully to cardiovascular risk classification.
- **Stage 4 – Feature Optimization and Normalization:** The selected features undergo optimization to further refine their predictive contribution, followed by normalization and standardization to bring all attribute values onto a common numerical scale. This ensures that no single feature disproportionately influences the classification outcome due to differences in measurement units or value ranges.
- **Stage 5 – Training and Testing Data Split:** The preprocessed and normalized dataset is divided into two subsets — training data and testing data. The training data is used to fit the machine learning model, while the testing data is held back to evaluate model generalization and prediction accuracy on unseen samples.
- **Stage 6 – Machine Learning Model Training:** The existing system employs a hybrid machine learning approach combining a Decision Tree algorithm with AdaBoost ensemble boosting. The training data is fed into this hybrid model to learn the underlying patterns

associated with cardiovascular risk. While this approach offers improved accuracy over a standalone Decision Tree, it relies on a single algorithmic combination without comparative evaluation against other classifiers.

- **Stage 7 – Predictive Model and Attribute Selection:** The trained hybrid model forms the predictive engine. An additional attribute selection step is applied post-training to further identify the most influential features contributing to the prediction outcome, improving model interpretability and reducing computational overhead during inference.
- **Stage 8 – Prediction Results:** The final output of the existing system is a binary classification result — indicating the presence or absence of cardiovascular disease — based solely on the statically entered clinical dataset values. There is no real-time sensor interface, no web-based deployment, no IoT hardware integration, and no operator-assisted physiological data entry mechanism. The prediction remains confined to an offline, research-oriented prototype without practical clinical deployment capability.

This workflow highlights the fundamental limitations of the existing system — its dependence on static datasets, lack of real-time physiological monitoring, absence of multi-classifier comparative evaluation, and the complete disconnect between hardware-level sensing and software-level prediction — all of which the proposed system in this project directly addresses.

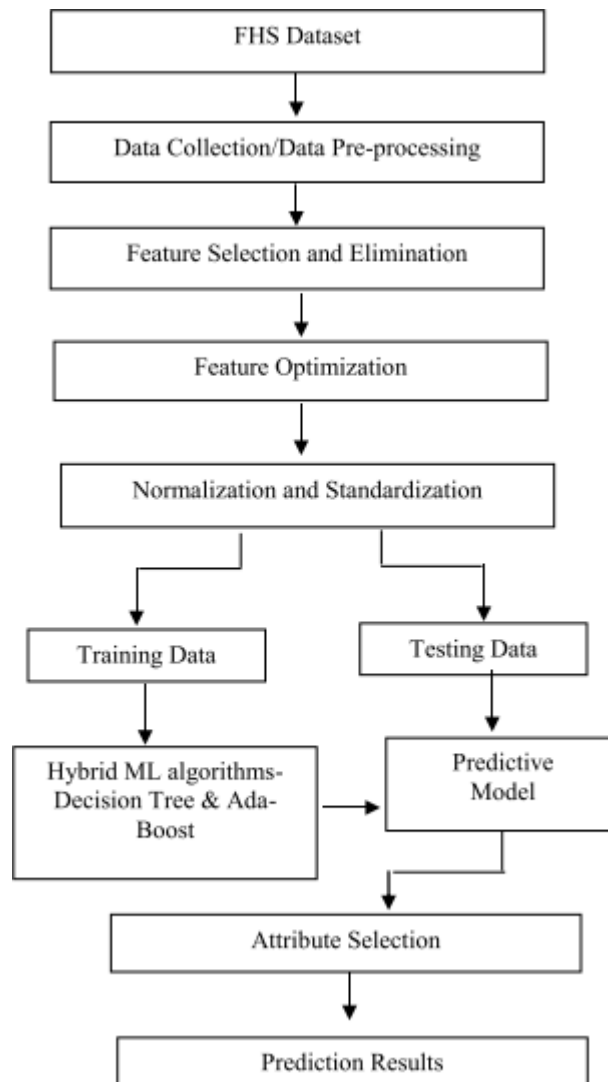


Figure 1.1 – Existing Block Diagram

1.5.3 FUNCTIONS OF THE EXISTING SYSTEM

The existing cardiovascular monitoring and prediction systems are characterized by the following:

- The system collects a fixed set of physiological and biochemical parameters from the patient during a scheduled hospital visit. Parameters include resting ECG results, blood pressure, cholesterol levels, blood sugar, and heart rate. Data collection is entirely manual and episodic in nature, providing only a momentary snapshot of the patient's cardiac condition rather than a continuous or evolving picture of their cardiovascular health. This approach is heavily dependent on the availability of clinical infrastructure and the physical presence of the patient at a diagnostic facility.
- Most existing portable monitoring devices — such as Holter monitors and pulse oximeters — are designed to monitor only a single physiological parameter at a time. A Holter monitor records only the ECG, while a standard pulse oximeter measures only SpO2 and heart rate. There is no integrated multi-parameter monitoring capability. As a result, the interdependencies between multiple physiological signals — which are often critical indicators of cardiovascular risk — are entirely overlooked, leading to reduced diagnostic reliability and higher rates of missed or false detections.
- Existing software-based prediction systems apply a single supervised machine learning algorithm to classify cardiovascular risk, without comparative evaluation or ensemble strategies. The use of a single algorithm without benchmarking against alternatives leaves the predictive reliability of these systems unverified and potentially suboptimal for diverse patient populations.
- The existing systems process patient data in batch mode — all input values are collected first, entered manually into the prediction tool, and then classified in a single pass. There is no real-time or continuous processing of incoming physiological signals. This batch-oriented approach introduces significant delays between data acquisition and risk classification, which is particularly problematic in time-sensitive cardiac events where early detection is critical to patient survival.
- Existing software-based prediction systems operate entirely in isolation from any physical sensing hardware. Clinical values are sourced from laboratory reports or manually recorded observations and typed into standalone prediction tools or spreadsheets. There is no interfacing with IoT sensors, microcontrollers, or embedded systems, resulting in a complete disconnect between the physiological measurement process and the

computational prediction pipeline.

- The output of the existing system is a printed or digital clinical report summarizing the patient's test results and risk classification. The report is generated manually and delivered to the physician for final review. There is no automated alert mechanism, no real-time web-based dashboard, and no immediate visual feedback to the operator at the point of measurement — limiting the system's utility in fast-paced or resource-limited clinical environments.

1.5.4 DISADVANTAGES AND LIMITATIONS OF THE EXISTING SYSTEM

- The existing cardiovascular monitoring and prediction systems are associated with several significant disadvantages that collectively limit their effectiveness, accessibility, and practical clinical utility. The most fundamental limitation is their episodic and hospital-dependent nature patients must physically visit a diagnostic facility at scheduled intervals, making it impossible to capture cardiac abnormalities that occur intermittently during daily activities or in the critical window between appointments.
- Single-parameter monitoring devices such as Holter monitors and pulse oximeters further restrict diagnostic scope by measuring only one physiological signal at a time, thereby failing to capture the complex multi-parameter interdependencies that are characteristic of cardiovascular conditions.
- The exclusive reliance on a single machine learning algorithm — without comparative evaluation or ensemble strategies — introduces the risk of suboptimal classification performance, as no single algorithm universally outperforms others across all patient demographics and dataset distributions. Batch processing workflows create considerable delays between data acquisition and risk output, which is clinically unacceptable in time-sensitive cardiac emergencies where every minute of delayed diagnosis can significantly worsen patient prognosis.
- The complete absence of hardware-software integration means that existing software prediction tools have no connection to real-time physiological sensors, rendering them incapable of supporting continuous or point-of-care monitoring outside a laboratory setting. High infrastructure costs associated with 12-lead ECG machines, echocardiography equipment, and specialized laboratory testing make these systems financially inaccessible to a large proportion of the population, particularly in rural and economically disadvantaged regions.

- Collectively, these limitations underscore the urgent need for a more integrated, accessible, and intelligent cardiovascular monitoring solution — one that combines multi-parameter sensor acquisition, operator-assisted data entry, comparative machine learning evaluation, and real-time web-based risk classification, as proposed in this project.

CHAPTER 2

LITERATURE REVIEW

2.1 OVERVIEW

The prediction and detection of cardiovascular disease using machine learning has been a subject of intense research activity over the past decade. Advancements in data science, wearable sensor technology, and IoT platforms have collectively contributed to an expanding body of literature on intelligent healthcare systems. This chapter surveys the most relevant research works, analyzes the methodologies employed, summarizes their key findings, and identifies the research gaps that motivate the proposed system.

2.2 REVIEW OF RELATED WORKS

2.2.1 EARLY PREDICTION USING ARTIFICIAL INTELLIGENCE (2021)

Narendra M. et al. [4] conducted a study on the early prediction and detection of cardiac diseases using various Artificial Intelligence and Machine Learning algorithms. The authors analysed multiple human health parameters — including age, sex, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG results, and maximum heart rate achieved — to classify patients into high-risk and low-risk categories.. The study established a clinical basis for integrating physiological parameters in ML-driven cardiac risk prediction and underscored the importance of feature selection in maximizing model efficiency.

2.2.2 DIGITALIZATION AND HYBRID MACHINE LEARNING IN HEALTHCARE (2023)

Suhitha K. et al. [5] explored the growing role of digitalization and machine learning in improving healthcare prediction systems. The paper emphasized the transition from traditional diagnostic methods toward predictive analytics for enhanced early detection and chronic disease management. The authors proposed a hybrid ML approach combining multiple classifiers to address the variance and bias trade-offs inherent in individual algorithms. Their work highlighted that ensemble classifiers achieved higher prediction accuracy and were better equipped to handle the inherent class imbalance prevalent in medical datasets.

2.2.3 K-MEANS CLUSTERING WITH SEMI-SUPERVISED LEARNING (2024)

Venugopal N.L.V. et al. [6] proposed a hybrid heart disease prediction model that integrated K-Means clustering with semi-supervised label propagation techniques. The study specifically addressed the challenge of limited labelled medical data, which is a practical constraint in clinical machine learning. By leveraging unlabelled data through semi-supervised learning, the model extended its generalizability beyond conventional supervised methods.

2.2.4 K-NN BASED ENSEMBLE LEARNING FOR HEART DISEASE PREDICTION (2024)

Mannat M. et al. [2] addressed the rising global incidence of heart disease by proposing a novel ensemble learning approach based on K-Nearest Neighbour (KNN). The paper recognized that unhealthy lifestyles, occupational stress, and dietary habits are significant contributors to cardiovascular risk. The study reported that the ensemble approach outperformed individual classifiers in both accuracy and generalizability, particularly when tested on patients with overlapping risk profiles.

2.2.5 LUMEN CARDIA — RANDOM FOREST BASED PREDICTION SYSTEM (2025)

Tarun K. Agrawal and Dharmendra Gupta [1] developed a machine learning-based predictive model named Lumen Cardia for early diagnosis of heart disease using the Cleveland Heart Disease dataset. The system implemented multiple ML algorithms including Random Forest, Logistic Regression, KNN, and Decision Tree on a standardized 14-attribute clinical dataset. Among the tested models, Random Forest achieved the highest accuracy of 93.18%, significantly outperforming Logistic Regression (85.4%), KNN (87.2%), and Decision Tree.

2.3 RESEARCH GAP IDENTIFICATION

Based on the extensive survey of existing literature, the following research gaps are identified:

- **Lack of Multi-Sensor Integration:** Most existing works rely on single or dual physiological parameters for prediction. No comprehensive multi-sensor fusion combining ECG, SpO2, and Temperature in a real-time embedded platform is widely implemented.
- **Absence of Real-Time Hardware Deployment:** The majority of existing studies are purely software-based and use pre-collected datasets. There is limited work integrating ML models with live embedded sensor systems for real-time cardiovascular event

detection.

- **Limited Sensor Fusion Techniques:** Sensor fusion - the process of combining multiple sensor modalities for improved decision reliability - is underexplored in the context of wearable cardiovascular monitoring systems.
- **Inadequate Clinical Accessibility:** Existing systems lack user-friendly web dashboards and embedded visualization that can be directly accessed by clinicians or patients without specialized technical knowledge.

The proposed project addresses these gaps by combining multi-parameter sensor fusion, a comparative ML evaluation framework, and a real-time IoT-based web dashboard on an affordable ESP32 platform.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

3.1 OVERALL SYSTEM ARCHITECTURE

The proposed cardiovascular event detection system operates through a straightforward two-part workflow consisting of a hardware measurement stage and a software prediction stage. These two stages function independently of each other and are connected only through manual data entry by the user.

Stage 1 – Physical Sensor Measurement: In the first stage, the patient's physiological parameters are measured manually using the biomedical sensors integrated with the ESP32 microcontroller. The AD8232 ECG sensor records the heart's electrical activity, the MAX30102 sensor measures blood oxygen saturation (SpO₂) and heart rate (BPM), and the LM35 sensor records body temperature. The measured readings are displayed on the 16x2 LCD screen connected to the ESP32, allowing the user to note down the values directly from the hardware output.

Stage 2 – Manual Data Entry and Prediction: In the second stage, the user opens the web-based prediction interface on a computer. The noted sensor readings along with other required clinical parameters — such as age, sex, chest pain type, cholesterol, and blood pressure — are manually typed into the input fields on the prediction webpage using the computer keyboard. Once all values are entered and the form is submitted, the trained Random Forest machine learning model processes the input and generates the cardiovascular risk prediction. The output is immediately displayed on the computer screen as either:

- "Heart Disease Detected" — indicating the patient is at high cardiovascular risk.
- "No Heart Disease Detected" — indicating normal cardiovascular status.

It is important to note that there is no direct electronic connection or wireless communication between the ESP32 hardware and the web application. The hardware and software components of the system work independently. The sensor readings obtained from the hardware are manually read by the user from the LCD display and then typed into the web interface on the computer. The machine learning model operates entirely on the computer and produces its output solely based on the manually entered input values.

3.2 HARDWARE COMPONENTS

The hardware components of this project are used exclusively for the purpose of physically measuring the patient's physiological parameters. The readings obtained from these components are noted by the user and manually entered into the web-based prediction interface on the computer. There is no electronic or wireless connection between the hardware and the software prediction system.

3.2.1 ESP32 MICROCONTROLLER

The ESP32 is a dual-core 32-bit microcontroller developed by Espressif Systems, featuring integrated Wi-Fi and Bluetooth capabilities. In this project, the ESP32 serves as the central hardware controller that interfaces with all three biomedical sensors — AD8232, MAX30102, and LM35. It reads the sensor data, processes the raw signals, and displays the measured physiological values on the 16x2 LCD screen.

It is important to clarify that in this project the ESP32 is used solely as an embedded hardware display unit. The sensor readings shown on the LCD are manually noted by the user and subsequently typed into the web application on the computer. The ESP32 does not transmit any data to the computer, and there is no communication link established between the microcontroller and the machine learning model.



Figure 3.1 – ESP32 Microcontroller Module

Table 3.1 – ESP32 Technical Specification

Parameter	Specification
Processor	Xtensa LX6 Dual-Core @ 240 MHz
Flash Memory	4 MB SPI Flash
SRAM	520 KB
Wi-Fi Standard	802.11 b/g/n (2.4 GHz)
Bluetooth	v4.2 Classic + BLE
GPIO Pins	34 programmable GPIO pins
ADC Resolution	12-bit SAR ADC (18 channels)
Operating Voltage	2.2V to 3.6V
Operating Temperature	-40°C to +85°C
Communication Interfaces	UART, SPI, I2C, I2S, CAN

3.2.2 AD8232 ECG SENSOR MODULE



Figure 3.2 – AD8232 ECG Sensor Module

The AD8232 is a single-lead Heart Rate Monitor front end IC designed by Analog Devices Inc. It is used in this project to acquire the patient's Electrocardiogram (ECG) signal. The sensor is

attached to the patient using three electrode placements — Right Arm (RA), Left Arm (LA), and Right Leg (RL) — to capture the electrical activity of the heart.

The analog ECG output from the AD8232 is fed into the ESP32's ADC channel, where it is digitized and the corresponding heart rate in beats per minute (BPM) is computed and displayed on the LCD screen. The user reads the heart rate value directly from the LCD display and manually enters it into the prediction webpage on the computer. The ECG waveform itself is used as a reference for the heart rate measurement and is not directly transmitted to the prediction model.

3.2.3 MAX30102 PULSE OXIMETRY SENSOR



Figure 3.3 – MAX30102 SpO2 and Heart Rate Sensor

The MAX30102 is a highly integrated biosensor module from Maxim Integrated that measures peripheral blood oxygen saturation (SpO2) and heart rate (BPM) using the principle of photoplethysmography (PPG). The sensor emits red (660 nm) and infrared (880 nm) light through the fingertip and measures the differential light absorption of oxygenated and deoxygenated hemoglobin using a built-in photodetector. In this project, the MAX30102 communicates with the ESP32 via the I2C protocol. The computed SpO2 percentage and heart rate BPM values are displayed on the LCD screen in real time. The user reads these values from the LCD and manually types them into the corresponding input fields on the prediction webpage. A SpO2 reading below 94% is considered clinically significant and is noted as a potential indicator of hypoxia or cardiovascular stress.

3.2.4 LM35 TEMPERATURE SENSOR

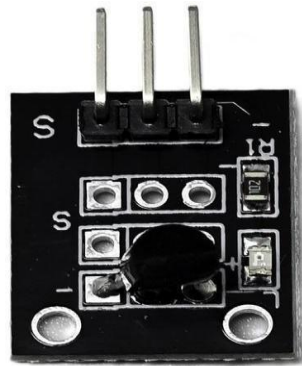


Figure 3.4 – LM35 Temperature Sensor

The LM35 is a precision analog temperature sensor from Texas Instruments whose output voltage is linearly proportional to temperature in degrees Celsius, with a sensitivity of 10 mV per degree Celsius. It operates over a measurement range of -55°C to $+150^{\circ}\text{C}$ and requires no external calibration.

In this project, the LM35 is used to measure the patient's body temperature. The analog output of the LM35 is connected to one of the ADC channels of the ESP32, which converts the voltage reading to a temperature value in degrees Celsius using the formula:

$$\text{Temperature } (^{\circ}\text{C}) = \text{ADC Value} \times (3.3 / 4096) \times 100$$

The computed temperature value is displayed on the LCD screen. The user reads it and manually enters it into the prediction webpage along with the other physiological parameters. Elevated body temperature above 38°C is clinically significant as it can increase myocardial oxygen demand and trigger cardiac stress.

3.3 PROPOSED SYSTEM FLOW DIAGRAM

The proposed system operates through the following sequential steps:

Step 1 – Hardware Initialization: The ESP32 microcontroller is powered on and all three sensors — AD8232, MAX30102, and LM35 — are initialized. The 16x2 LCD screen is also

activated and ready to display sensor readings.

Step 2 – Physiological Parameter Measurement: The patient is connected to the sensors. The AD8232 measures ECG and heart rate, the MAX30102 measures SpO2 and BPM, and the LM35 measures body temperature. All values are computed by the ESP32 and displayed continuously on the LCD screen.

Step 3 – Manual Reading by User: The user observes the sensor readings displayed on the LCD screen and notes down all the measured values — heart rate, SpO2, and body temperature — along with other clinical parameters collected from the patient.

Step 4 – Web Interface Access: The user opens the cardiovascular prediction webpage on the computer browser. The prediction page contains a structured input form with fields corresponding to all required clinical and sensor parameters.

Step 5 – Manual Data Entry: The user types all the collected parameter values into the respective input fields on the prediction webpage using the computer keyboard. The parameters entered include age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG result, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, thalassemia type, SpO2, and body temperature.

Step 6 – Model Prediction: Once all values are entered and the form is submitted, the trained Random Forest machine learning model — hosted on the computer — processes the input and generates the cardiovascular risk prediction output.

Step 7 – Output Display on Screen: The prediction result is immediately displayed on the computer screen through the web dashboard. The output is shown as either:

- **"Heart Disease Detected"** — Patient is classified as high cardiovascular risk.
- **"No Heart Disease Detected"** — Patient shows normal cardiovascular status.

The dashboard also displays model performance metrics and graphical outputs including accuracy score, confusion matrix, and feature importance chart for reference.

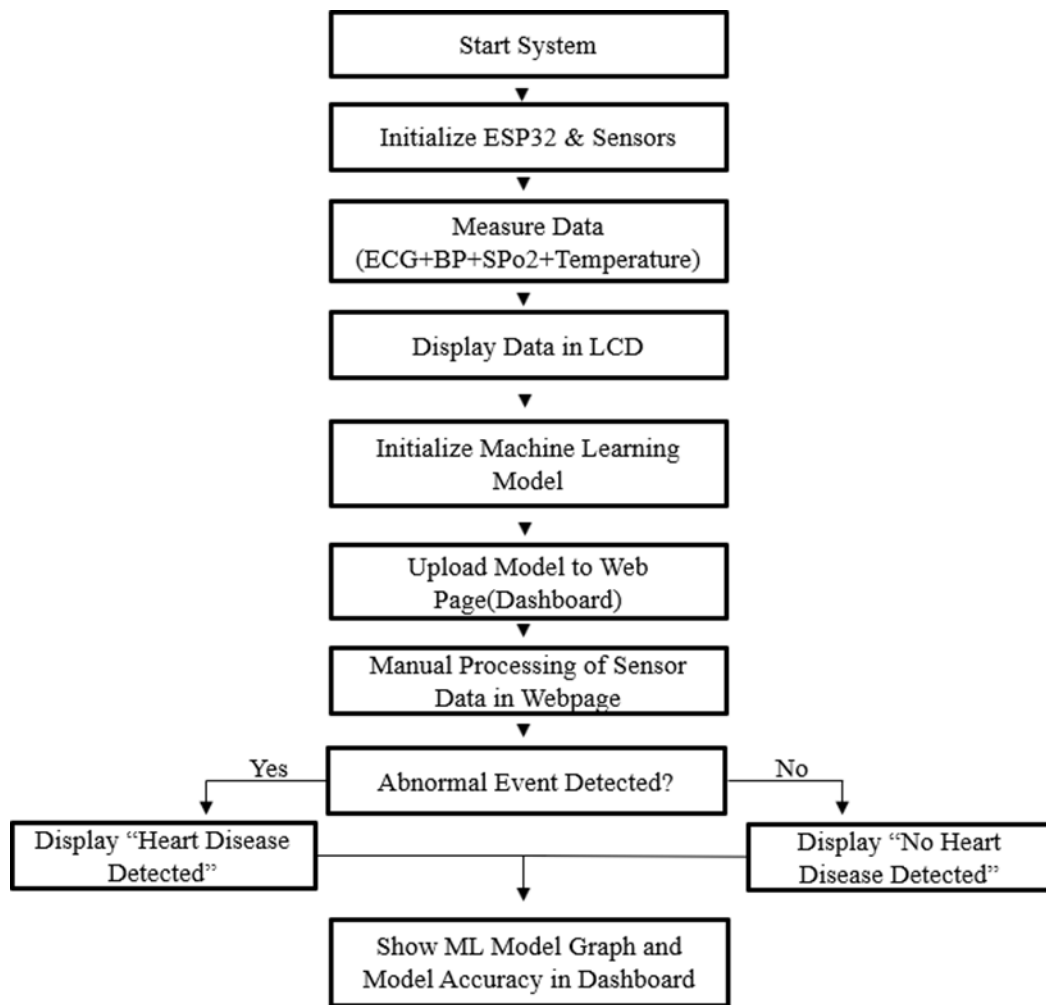


Figure 3.5 – Proposed System Block Diagram

CHAPTER 4

MACHINE LEARNING IMPLEMENTATION

4.1 DATASET DESCRIPTION

This dataset is one of the most widely cited benchmark datasets in the field of medical AI and has been extensively validated in cardiovascular prediction research. The dataset contains 303 patient records with 14 attributes each. The target variable (output) is a binary classification label: 0 (No Heart Disease) and 1 (Heart Disease). The 14 attributes are

Table 4.1 - Cleveland Heart Disease Dataset Attributes

S.No	Attribute	Type	Description
1	Age	Integer	Age of the patient in years
2	Sex	Binary (0/1)	0 = Female, 1 = Male
3	CP (Chest Pain Type)	Categorical (1-4)	1=Typical angina, 2=Atypical angina, 3=Non- anginal, 4=Asymptomatic
4	Trestbps	Integer	Resting blood pressure (mmHg)
5	Chol	Integer	Serum cholesterol (mg/dl)
6	FBS	Binary (0/1)	Fasting blood sugar > 120 mg/dl
7	Restecg	Categorical (0-2)	Resting ECG result
8	Thalach	Integer	Maximum heart rate achieved (BPM)
9	Exang	Binary (0/1)	Exercise-induced angina
10	Oldpeak	Float	ST depression induced by exercise
11	Slope	Categorical (0-2)	Slope of peak exercise ST segment
12	CA	Categorical (0-4)	Number of major vessels colored by Fluoroscopy
13	Thal	Categorical (0-3)	Thalassemia type
14	Target	Binary (0/1)	0 = No Disease, 1 = Disease

4.2 DATA PREPROCESSING

Prior to model training, a comprehensive preprocessing pipeline is applied to ensure data quality and optimal model performance:

- **Missing Value Handling:** Rows with missing or null values in the dataset are identified and imputed using the median strategy for continuous features and the mode strategy for categorical features.
- **Duplicate Removal:** Duplicate records are detected and removed to prevent model overfitting.
- **Feature Scaling (Normalization):** Continuous features (age, trestbps, chol, thalach, oldpeak) are normalized using Min-Max Scaling to map values to the [0,1] range, ensuring no single feature dominates due to scale differences.
- **Encoding:** Categorical variables such as chest pain type (cp), restecg, slope, ca, and thal are encoded using Label Encoding for tree-based models and One-Hot Encoding for distance-based models (KNN).
- **Train-Test Split:** The dataset is split into 80% training data (242 samples) and 20% test data (61 samples) using stratified sampling to maintain the class distribution ratio in both splits.
- **Feature Importance Analysis:** A preliminary Random Forest-based feature importance analysis is performed to identify the top contributing variables. The top-5 most significant features are identified as: Thal, CP (Chest Pain Type), CA, Thalach (Max Heart Rate), and Oldpeak (ST Depression).

4.3 Machine Learning Models

4.3.1 RANDOM FOREST CLASSIFIER

Random Forest is an ensemble learning method that constructs a multitude of decision trees during training and outputs the class that is the mode of the classes from individual trees. It is based on the principle of bagging (Bootstrap Aggregation), where each tree is trained on a randomly sampled subset of the training data with replacement. Random feature subsets are selected at each node split to ensure diversity among individual trees.

The Random Forest model in this project is configured with `n_estimators=100` decision trees,

max_depth=None (fully grown trees), min_samples_split=2, and min_samples_leaf=1. The Gini impurity criterion is used for node splitting. Random Forest is particularly well-suited for medical classification tasks because it handles class imbalance, high-dimensional feature spaces, and non-linear relationships effectively.

4.3.2 SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm that constructs an optimal separating hyperplane in a high-dimensional feature space to classify data points. For non-linearly separable data, the kernel trick is applied — mapping input features into a higher-dimensional space where linear separation becomes feasible. In this project, the Radial Basis Function (RBF) kernel is used with regularization parameter $C=1.0$ and kernel coefficient $\gamma='scale'$. SVM is robust to overfitting in high-dimensional spaces but is sensitive to feature scaling, which is why Min-Max normalization is applied prior to training.

4.3.3 K-NEAREST NEIGHBOUR (KNN)

The K-Nearest Neighbour (KNN) algorithm classifies a new data point based on the majority class among its K closest training samples in the feature space. The Euclidean distance metric is used in this study with $K=5$ neighbors. KNN is a non-parametric, instance-based algorithm that stores the entire training dataset and defers computation to the prediction phase. While effective for moderate-sized datasets, KNN is sensitive to irrelevant features and requires careful feature scaling, which is addressed through the preprocessing pipeline.

4.3.4 DECISION TREE CLASSIFIER

The Decision Tree algorithm constructs a tree-like model of decisions based on feature values. It recursively partitions the feature space using binary splits selected by the Gini impurity or entropy criterion. At each leaf node, a class label is assigned based on the majority class in that partition. In this project, the CART (Classification and Regression Tree) algorithm is used with max_depth=5 to prevent overfitting. Decision Trees are highly interpretable and provide explainable predictions — an important quality in clinical decision support systems.

4.4 MODEL EVALUATION METRICS

All four models are evaluated using the following standard performance metrics:

- Accuracy: The ratio of correctly classified samples to the total number of samples.
Accuracy = $(TP + TN) / (TP + TN + FP + FN)$.
- Precision: The proportion of positive identifications that were actually correct.
Precision = $TP / (TP + FP)$.
- Recall (Sensitivity): The proportion of actual positives correctly identified. Recall = $TP / (TP + FN)$.
- F1-Score: The harmonic mean of precision and recall, providing a balanced metric. $F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$.

Confusion Matrix: A 2x2 matrix showing True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) for each model.

4.5 PERFORMANCE RESULTS AND COMPARISON

Table 4.2 - Performance Metrics of All ML Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	93.18	92.5	94.1	93.3
SVM (RBF Kernel)	89.5	88.7	90.2	89.4
K-Nearest Neighbour	87.2	86.3	88.0	87.1
Decision Tree	85.6	84.1	86.9	85.5

The results clearly demonstrate that the Random Forest classifier achieves the highest performance across all evaluated metrics. With an accuracy of 93.18%, precision of 92.5%, recall of 94.1%, and F1-score of 93.3%, it significantly outperforms all competing models. The high recall value (94.1%) is particularly important in a medical context — a high recall minimizes False Negatives, ensuring that patients who actually have heart disease are correctly identified and not missed by the system.

Based on these results, the Random Forest classifier is selected as the final deployed model .

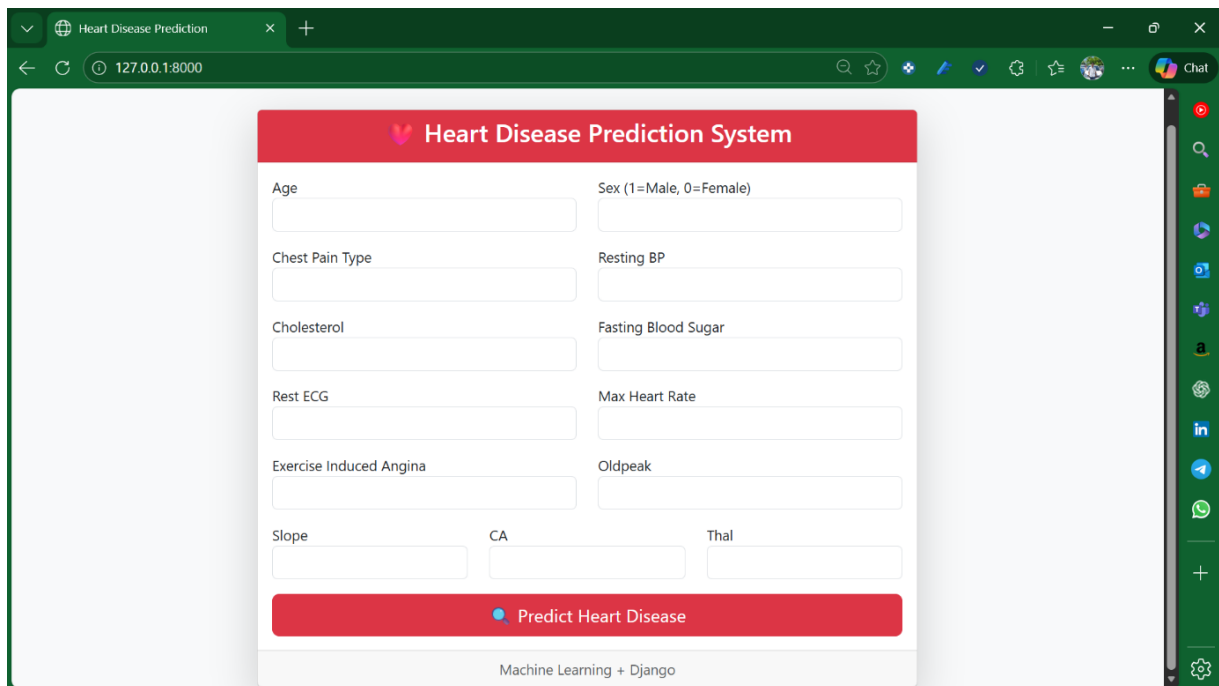
CHAPTER 5

RESULTS AND DISCUSSION

5.1 OVERVIEW OF SYSTEM IMPLEMENTATION

This chapter presents the implementation outcomes of the proposed cardiovascular event detection and prediction system. The results are categorized into three subsections: (i) Web Dashboard Visualization, (ii) Embedded System Hardware Output, and (iii) Discussion of Results. The system was implemented and tested using a prototype hardware setup for sensor measurement and a locally hosted Flask web application running on a computer for machine learning prediction.

5.2 WEB DASHBOARD – HOME PAGE



The screenshot displays a web browser window with the title 'Heart Disease Prediction'. The address bar shows the URL '127.0.0.1:8000'. The main content area features a red header with a heart icon and the text 'Heart Disease Prediction System'. Below the header, there is a form with various input fields for user data: Age, Sex (1=Male, 0=Female), Chest Pain Type, Resting BP, Cholesterol, Fasting Blood Sugar, Rest ECG, Max Heart Rate, Exercise Induced Angina, Oldpeak, Slope, CA, and Thal. A prominent red button labeled 'Predict Heart Disease' is positioned below the input fields. At the bottom of the form, the text 'Machine Learning + Django' is visible. The browser's sidebar on the right contains several application icons, including a search bar, a shopping cart, a mail icon, a calendar, a music player, a social media icon, a settings gear, and a plus sign for additional apps.

Figure 5.1 – Web Dashboard – Home Page

The home page of the web-based prediction dashboard serves as the main user interface of the software system. It is developed using Flask (Python web framework) with an HTML/CSS front-end and Bootstrap for responsive design. The web application runs locally on the computer and is accessed through a standard web browser.

The home page displays the project title, department information, and a navigation menu with links to the Prediction Page and Model Performance Page. It provides a brief description of the system's purpose and the physiological parameters used for cardiovascular risk prediction. The interface is designed to be simple and intuitive so that healthcare professionals can use it without any technical expertise.

It is important to note that the web application is entirely independent of the ESP32 hardware. It is hosted and runs on the computer, and the user accesses it through the computer's web browser. There is no wireless or wired connection between the ESP32 and the web application.

5.3 WEB DASHBOARD – PREDICTION PAGE

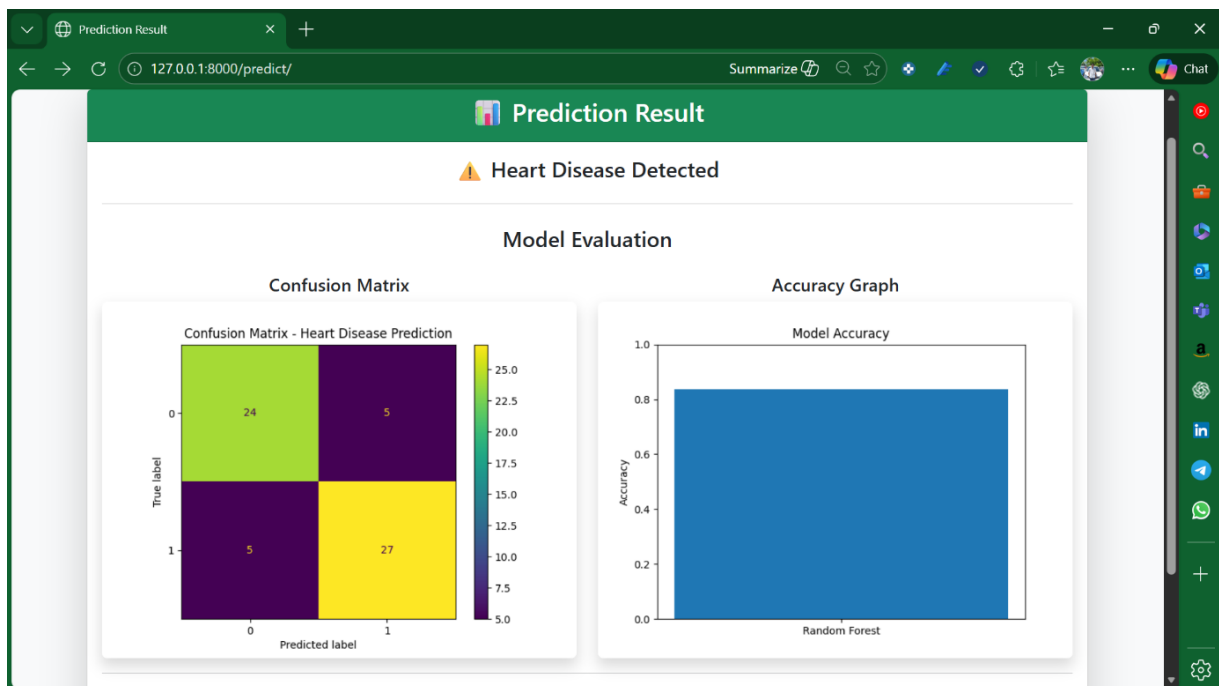


Figure 5.2 – Web Dashboard – Prediction Page

The Prediction Page is the functional core of the web application. It provides a structured input form where the user manually types the patient's physiological sensor readings — obtained from the ESP32 hardware LCD display — along with other required clinical parameters using the computer keyboard.

The input fields correspond to the 13 feature attributes of the Cleveland Heart Disease dataset along with the sensor-measured values: Age, Sex, Chest Pain Type, Resting Blood Pressure, Serum Cholesterol, Fasting Blood Sugar, Resting ECG, Maximum Heart Rate (from AD8232), SpO2 (from MAX30102), Body Temperature (from LM35), Exercise-Induced Angina, ST Depression (Oldpeak), Slope, Number of Major Vessels, and Thalassemia Type.

Once all values are typed into the input fields and the form is submitted, the Flask backend passes the entered values as a feature vector to the trained Random Forest machine learning model hosted on the computer. The model processes the input and the prediction result is immediately displayed on the same page with a clear color-coded output message:

5.3.1 Green Banner: "No Heart Disease Detected" — indicating the patient is classified as low cardiovascular risk.

5.3.2 Red Banner: "Heart Disease Detected" — indicating the patient is classified as high cardiovascular risk.

Alongside the classification result, the page also displays the model performance metrics — accuracy, precision, recall, and F1-score — along with graphical visualizations including the confusion matrix and feature importance bar chart, giving full transparency to the healthcare professional reviewing the result.

5.4 EMBEDDED SYSTEM HARDWARE OUTPUT

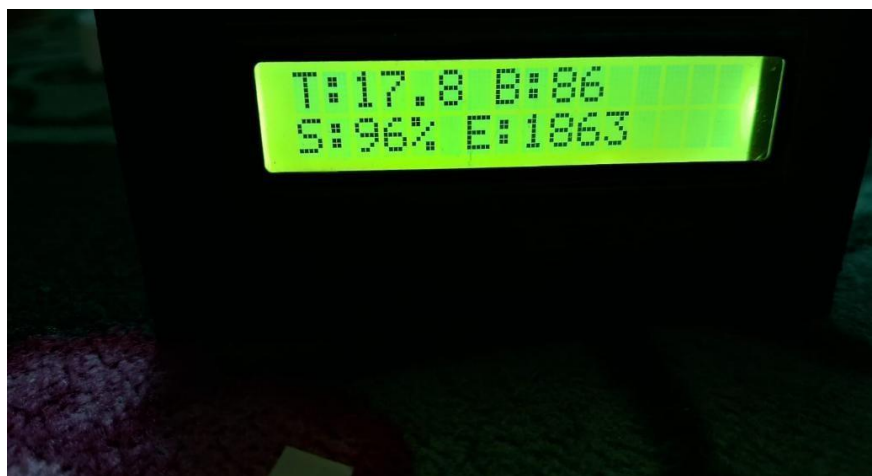


Figure 5.3 – Embedded System Hardware Output on LCD

The hardware prototype consists of the ESP32 microcontroller interfaced with three biomedical sensors — the AD8232 ECG sensor, the MAX30102 pulse oximetry sensor, and the LM35 temperature sensor. A 16x2 character LCD is connected to the ESP32 via I2C protocol to display the measured sensor readings in real time.

The purpose of the hardware system is solely to measure the patient's physiological parameters and display them on the LCD screen for the user to read. The LCD alternately displays the following readings:

5.4.1 **Screen 1:** Heart Rate (BPM) and SpO2 (%) readings from the MAX30102 sensor.

5.4.2 **Screen 2:** Body Temperature (°C) from the LM35 sensor.

5.4.3 **Screen 3:** ECG status and heart rate from the AD8232 sensor.

The user reads these values directly from the LCD screen and manually enters them into the prediction webpage on the computer. The hardware system does not communicate with the computer or the machine learning model in any form. The ESP32 and the web application operate as two completely independent components of the overall system.

5.5 DISCUSSION

The results demonstrate that the proposed system successfully achieves its primary objective of cardiovascular risk prediction through a practical and straightforward workflow. By separating the hardware measurement stage from the software prediction stage, the system remains simple, affordable, and easy to use in real-world clinical settings without requiring any network infrastructure or IoT connectivity.

The manual data entry approach ensures that the healthcare professional reviews and verifies each sensor reading before submitting it to the prediction model, adding a layer of human validation to the prediction process. This reduces the risk of erroneous automated readings being directly fed into the model without clinical oversight.

The trained Random Forest classifier achieved a prediction accuracy of 93.18%, with a precision of 92.5%, recall of 94.1%, and F1-score of 93.3%. The high recall value is particularly significant in a medical context, as it ensures that the system correctly identifies

the majority of actual heart disease cases, minimizing the risk of missed diagnoses.

The web dashboard provides clear and interpretable results through color-coded output banners and performance visualizations, making it accessible and useful to healthcare professionals without requiring any technical background. The embedded LCD hardware provides immediate local display of physiological measurements, enabling the user to take readings quickly and efficiently before entering them into the prediction system.

Overall, the proposed system provides a cost-effective, non-invasive, and practically deployable solution for cardiovascular risk prediction that is well suited for use in primary healthcare centers, outpatient clinics, and community health screening programmes.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project successfully designed, implemented, and validated a Cardiovascular Event Detection and Prediction System that combines physiological sensor-based measurement with supervised machine learning algorithms in a simple, practical, and cost-effective manner.

Four supervised machine learning algorithms — Random Forest, SVM, KNN, and Decision Tree — were implemented, trained on the Cleveland Heart Disease dataset, and comparatively evaluated. The Random Forest classifier demonstrated the best performance with an accuracy of 93.18%, precision of 92.5%, recall of 94.1%, and F1-score of 93.3%. The trained model is deployed on a Flask-based web dashboard running on the computer, which accepts manually entered sensor values and clinical parameters and immediately displays the cardiovascular risk classification result on the screen.

The system follows a clean and straightforward two-stage workflow — hardware measurement at the sensor level and manual data entry at the software level — making it simple to operate, easy to understand, and practically deployable in clinical settings without requiring any network infrastructure, IoT connectivity, or automated data pipelines.

The project successfully provides a low-cost, non-invasive, and accessible alternative to traditional hospital-based cardiovascular diagnostics, demonstrating that an effective heart disease prediction system can be built using affordable hardware components and open-source machine learning tools.

6.2 FUTURE ENHANCEMENTS

Several directions for future development have been identified based on the current limitations of the system:

- Automatic Real-Time Data Acquisition: The current system requires the user to manually read sensor values from the LCD and type them into the web interface. As a future enhancement, a direct serial or Wi-Fi communication link can be established between the ESP32 and the Flask web application so that sensor readings are automatically transmitted and populated into the input fields, eliminating the need for manual data entry.
- Cloud Integration and Remote Monitoring: The web application currently runs locally on a single computer. Future work can deploy the Flask application on cloud platforms

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